

1. AUTO_INCREMENT

`department_id INT NOT NULL AUTO_INCREMENT`

What it does in this table:

- Automatically generates a new integer value for `department_id` whenever you insert a new row into `parks_departments` and do not specify `department_id` explicitly.
- Starts at 1 by default and increases by 1 for each new department.
- Ensures each department gets a unique ID without you having to manually manage the numbers.

Why it is useful here:

- `department_id` is the primary key for `parks_departments`, so it must be unique for every row.
- **AUTO_INCREMENT:**
 - Avoids accidental duplication of IDs.
 - Simplifies inserts like:

```
INSERT INTO parks_departments (department_name)
VALUES ('Parks and Recreation');
```

You don't have to write the `department_id`; it is filled in automatically.

- This makes `dept_id` in `employee_salary` easy to maintain as a reference to `parks_departments.department_id` (even though you did not declare a formal foreign key).

2. NOT NULL

You use NOT NULL on:

- `employee_demographics.employee_id INT NOT NULL`
- `employee_salary.employee_id INT NOT NULL`
- `employee_salary.first_name VARCHAR(50) NOT NULL`
- `employee_salary.last_name VARCHAR(50) NOT NULL`
- `parks_departments.department_id INT NOT NULL AUTO_INCREMENT`

- `parks_departments.department_name VARCHAR(50) NOT NULL`

What NOT NULL does:

- Forbids the column from having a `NULL` value (unknown/missing).
- The database will reject any `INSERT` or `UPDATE` that tries to leave this column empty.

Why each use makes sense:

- `employee_id INT NOT NULL`
 - Every employee must have an ID; without it you cannot reliably identify or join records.
 - Being `PRIMARY KEY` in `employee_demographics` also implicitly enforces `NOT NULL`.
- `first_name VARCHAR(50) NOT NULL, last_name VARCHAR(50) NOT NULL` in `employee_salary`
 - Ensures that salary records always have clear name information.
 - Prevents “nameless” salary rows that would be confusing and hard to match.
- `department_id INT NOT NULL AUTO_INCREMENT`
 - As a primary key it must always be present; `NOT NULL` ensures you never have a department without an ID.
 - Works together with `AUTO_INCREMENT`: the system must always generate a valid value, not `NULL`.
- `department_name VARCHAR(50) NOT NULL`
 - Every department must have a name; a department with an ID but no name would be meaningless for users and reports.

How they work together

- In `parks_departments`, `department_id INT NOT NULL AUTO_INCREMENT + PRIMARY KEY` means:
 - The database automatically assigns a unique, non-null ID for each new department.
 - You cannot insert duplicate or missing IDs.
- In other tables, `NOT NULL`:

- Guarantees that critical identifying fields (IDs, names) are always present.
- Improves data quality and makes joins and queries more reliable.